

FEETECH Continuous Rotation Servo FS5106R

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The FS5106R is a standard-sized servo that has been built by FEETECH specifically for continuous rotation, making it an easy way to get your robot moving. It features two ball bearings on the output shaft for reduced friction, and it offers easy access to the rest-point adjustment potentiometer. This servo can work with both 5 V and 3.3 V servo signals.

Key specs at 6 V: 95 RPM (no-load), 83 oz-in (6 kg-cm), 43 g, CCW rotation for pulse widths above the rest point.

or .

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Overview

The FS5106R is a standard-sized servo that has been built by FEETECH (formerly known as Fitec) specifically for continuous rotation. At 6 V, it has a maximum rotation speed of 95 RPM (no-load) and can produce up to 83 oz-in (6 kg-cm) of torque. It features two ball bearings on the output shaft for reduced friction, and it offers easy access to the rest-point adjustment potentiometer. The servo can be controlled using a direct connection to a single microcontroller I/O line without any additional electronics, which makes it a great actuator for beginner robotics projects.

The FS5106R continuous rotation servo converts standard RC servo position pulses into continuous rotation speed. The default rest point is 1.5 ms, but this can be adjusted by using a small screwdriver to turn the middle-point positioner. Pulse widths above the rest point result in counterclockwise rotation, with speed increasing as the pulse width increases; pulse widths below the rest point result in clockwise rotation, with speed increasing as the pulse width decreases.

The servo has a 12" (300 mm) lead that is terminated with a JR-style connector and includes additional servo horns and mounting hardware. The following picture shows an example of the hardware that might be included (actual hardware could vary):



This robotics servo is compatible with our [servo controllers](#) and has a Futaba-compatible output shaft, which means it works with our [servo wheels](#) for standard servos (25T, 5.8 mm spline). It is also compatible with our [Mounting Bracket for Standard-Size Servos](#):



You can find more information about this servo in its [datasheet](#) (463k pdf) (please note that the free-run speed specifications appear to be incorrect in this datasheet; in our tests, the maximum no-load speeds were 95 RPM at 6 V and 78 RPM at 4.8 V, and these are the numbers we have published on our product page).

Note that, as with most hobby servos, stalling or back-driving this servo can damage it.

FS5106R vs FS5106B

Two versions of this servo are available. This product is the FS5106R, which offers open-loop speed control over an unlimited range. Alternatively, the [FS5106B](#) offers closed-loop position control over a limited range (approximately 180°). The FS5106R and FS5106B have identical dimensions and use the same motor and gear reduction.

Continuous rotation servo comparison



Continuous rotation servo size comparison. From left to right: SpringRC SM-S4303R, Power HD AR-3606HB, FEETECH FS5106R, Parallax Feedback 360°, Parallax (Futaba S148), and FEETECH FS90R.

	6 V		4.8 V		Weight (g)	Size (mm)	Direction vs pulse (1ms → 2ms)	Digital?	Feed-back?	Price
	Max speed (RPM)	Stall torque (oz·in)	Max speed (RPM)	Stall torque (oz·in)						
<u>SpringRC SM-S4303R</u>	54	71	43	46	41	41.3 × 20.7 × 40.2	CW→CCW			\$21.39
<u>SURPLUS SpringRC SM-S4303R 2-Pack</u>	54	71	43	46	41	41.3 × 20.7 × 40.2	CCW→CW			\$24.74 for a pair of servos
<u>FEETECH FS5106R</u>	95	83	75	67	43	40.8 × 20.1 × 38.0	CW→CCW			\$24.35
<u>Parallax Feedback 360°</u>	140	35	–	–	41	40.0 × 20.0 × 37.2	CW→CCW	✓	✓	\$29.95
<u>FEETECH FT90R</u>	135	21	108	18	9	22.9 × 12.1 × 26.5	CW→CCW	✓		\$12.63
<u>FEETECH FS90R</u>	130	21	100	18	9	22.5 × 12.1 × 23.4	CW→CCW			\$9.82